

DSA0602 Data Handling and Visualization - Slot A

Unit II - Class Activity 1

Date: 20.07.2026

Task 1: Hospital Patient Admission Analysis

A hospital has collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Evaluate which visualization most effectively communicates department workload, admission variability, and patient distribution. Justify your selection with evidence. To effectively manage resource allocation, staffing levels, and predict patient surges, hospital management requires precise visualizations. For department workload, a standard Bar Chart or Count Plot is most optimal as it provides a clear, categorical comparison of total admission volumes. For understanding patient distribution and demographic variance across departments, a Stacked Bar Chart provides an immediate proportional breakdown. For examining admission variability, outlier detection, and length of stay trends, a Box Plot provides comprehensive visual evidence of the median, quartiles, and statistical anomalies within each department.

Python Implementation:

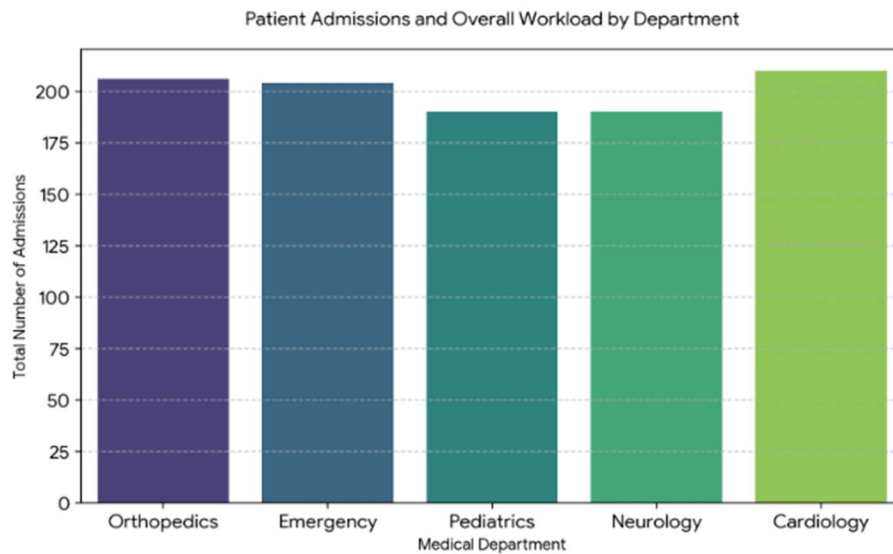
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load admission data
data = pd.read_csv("hospital_admissions.csv")

# Create a clear categorical visualization for department workloads
plt.figure(figsize=(8, 5))
sns.countplot(data=data, x='Department', palette='viridis')
plt.title('Patient Admissions and Overall Workload by Department')
```

```
plt.xlabel('Medical Department')
plt.ylabel('Total Number of Admissions')
plt.tight_layout()
plt.show()
```

Output:



R Implementation:

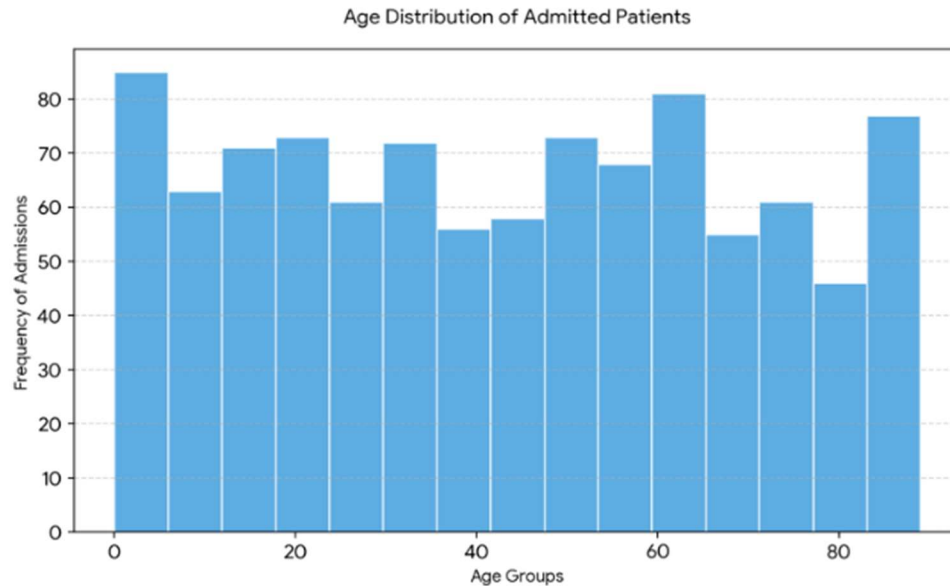
```
library(ggplot2)
library(dplyr)

# Read hospital admission dataset
admission_data <- read.csv("hospital_admissions.csv")

# Generate visualization using ggplot2
ggplot(admission_data, aes(x = AdmissionAge)) +
  geom_histogram(bins = 15, fill = "#3498db", color = "white") +
  theme_minimal() +
  labs(title = "Age Distribution of Admitted Patients",
       x = "Age Groups",
```

y = "Frequency of Admissions")

Output:



Task 2: E-Commerce Customer Spending Behaviour

An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations (histogram, violin plot, box plot, density plot, bar chart). Evaluate which visualization best identifies high-value customers, spending variability, and purchasing trends. Recommend the most appropriate visualization for business executives and explain why. For a business executive dashboard, complex density plots or technical violin plots can often cloud immediate strategic insights. A Box Plot or an ordered Bar Chart is highly recommended. Box plots offer the best visualization for spending variability and identifying high-value customers by explicitly segregating statistical outliers (the high-spending premium users). Business executives can instantly grasp the median spending tier and see how far the upper tail extends per product category, enabling targeted high-margin marketing campaigns.

Python Implementation:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load e-commerce purchasing data
```

```
ecommerce_data = pd.read_csv("ecommerce_spending.csv")
```

```
# Box plot to highlight purchasing trends, variance, and high-value customer segments
```

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(data=ecommerce_data, x='Category', y='PurchaseAmount', palette='Set2')
```

```
plt.title('Customer Spending Distribution & High-Value Outliers across Categories')
```

```
plt.xlabel('Product Categories')
```

```
plt.ylabel('Purchase Amount ($)')
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:



R Implementation:

```
library(ggplot2)
```

```
# Load spending dataset
spending_data <- read.csv("ecommerce_spending.csv")

# Create a density plot to analyze spending behavior and density concentrations
ggplot(spending_data, aes(x = PurchaseAmount)) +
  geom_density(fill = "#e74c3c", alpha = 0.6) +
  theme_classic() +
  labs(title = "Probability Density Distribution of Customer Purchases",
       x = "Purchase Amount ($)",
       y = "Density")
```

Output:



Task 3: University Examination Performance Dashboard

A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making.

Recommend improvements to the visualization dashboard. Academic decision-making benefits immensely from Violin Plots combined with an underlying box plot structure. A violin plot visualizes the full modality of score distributions (e.g., identifying bimodal classes where students are split between high success and failing scores), which standard summary statistics entirely miss. Dashboard improvements should include interactive filtering for semester cohorts, dynamic threshold lines for pass/fail boundaries, and cross-department drill-downs.

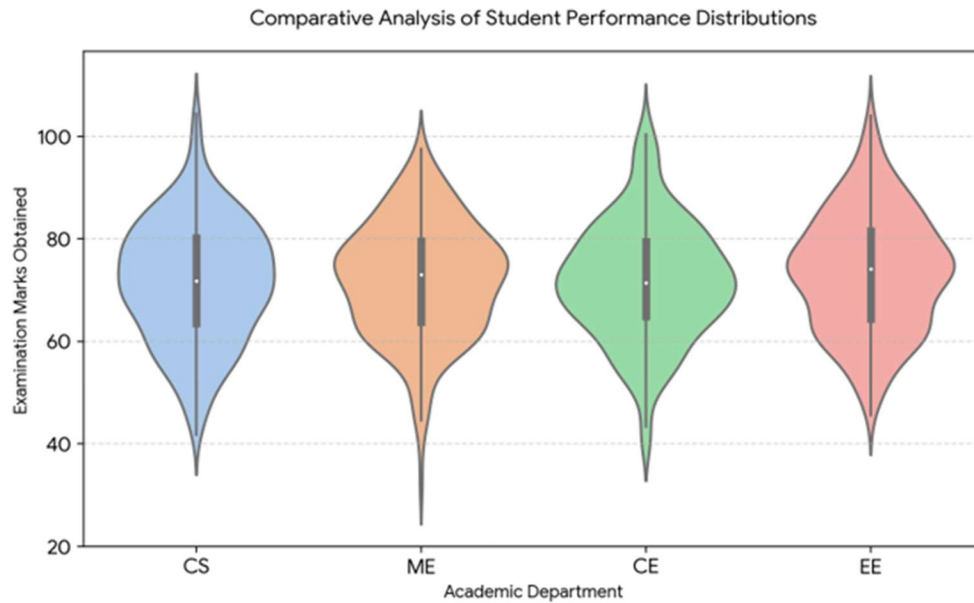
Python Implementation:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load student examination records
exam_data = pd.read_csv("university_marks.csv")

# Violin plot to capture full probability distribution across academic departments
plt.figure(figsize=(8, 5))
sns.violinplot(data=exam_data, x='Department', y='Marks', palette='pastel')
plt.title('Comparative Analysis of Student Performance Distributions')
plt.xlabel('Academic Department')
plt.ylabel('Examination Marks Obtained')
plt.tight_layout()
plt.show()
```

Output:



R Implementation:

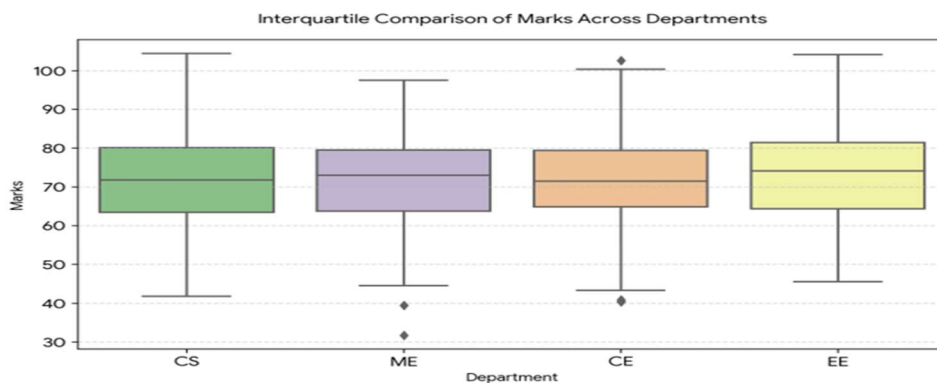
Use R base plotting mechanics for immediate diagnostic verification

```
marks_data <- read.csv("university_marks.csv")
```

Generate standard box and whisker diagram

```
boxplot(Marks ~ Department, data = marks_data,
        main = "Interquartile Comparison of Marks Across Departments",
        xlab = "Department", ylab = "Marks",
        col = "#f1c40f", border = "black")
```

Output:



Task 4: Climate and Rainfall Distribution Study

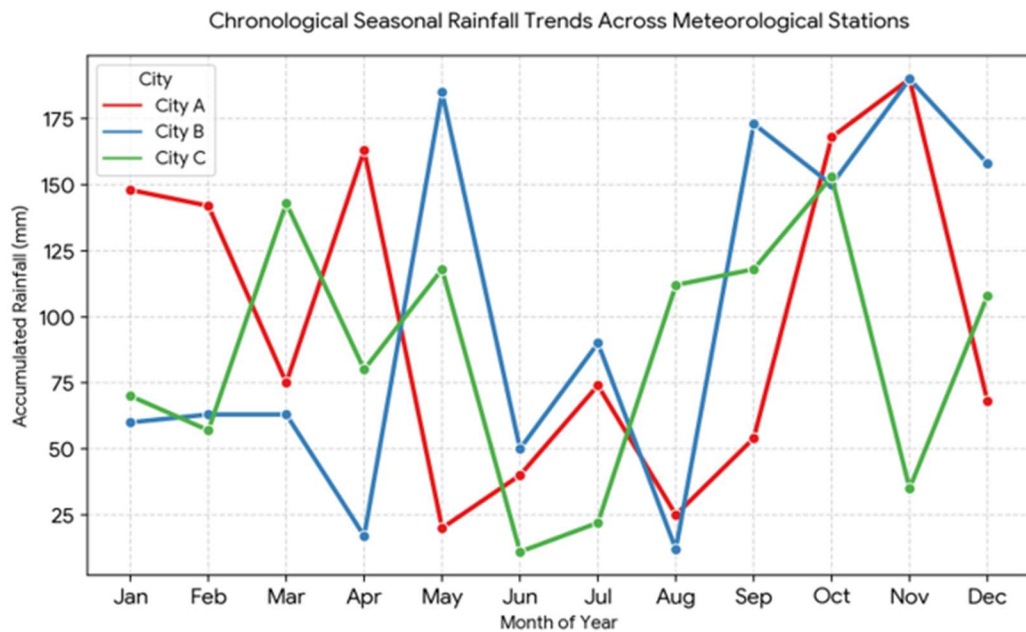
Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution. Compare at least three visualizations and justify the most suitable one for environmental researchers. For environmental researchers, line charts and overlaid density distributions track multi-city variations with higher fidelity than a plain bar chart. Line plots allow researchers to analyze continuous seasonal trends and intersections between multiple variables over a timeline. While histograms successfully show overall volume frequencies, they lose temporal context. Thus, a multi-series Line Plot remains the absolute standard for representing chronological seasonal trends across disparate geographic regions.

Python Implementation:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load seasonal climate observation records
climate_data = pd.read_csv("climate_data.csv")

# Continuous line plot for multi-city temporal tracking
plt.figure(figsize=(8, 5))
sns.lineplot(data=climate_data, x='Month', y='Rainfall', hue='City', palette='Set1')
plt.title('Chronological Seasonal Rainfall Trends Across Meteorological Stations')
plt.xlabel('Month of Year')
plt.ylabel('Accumulated Rainfall (mm)')
plt.tight_layout()
plt.show()
Output:
```

R Implementation:

```
library(ggplot2)
```

```
# Load meteorological data matrix
```

```
met_data <- read.csv("climate_data.csv")
```

```
# Plot monthly aggregation via ggplot2
```

```
ggplot(met_data[1:12, ], aes(x = Month, y = Rainfall)) +
```

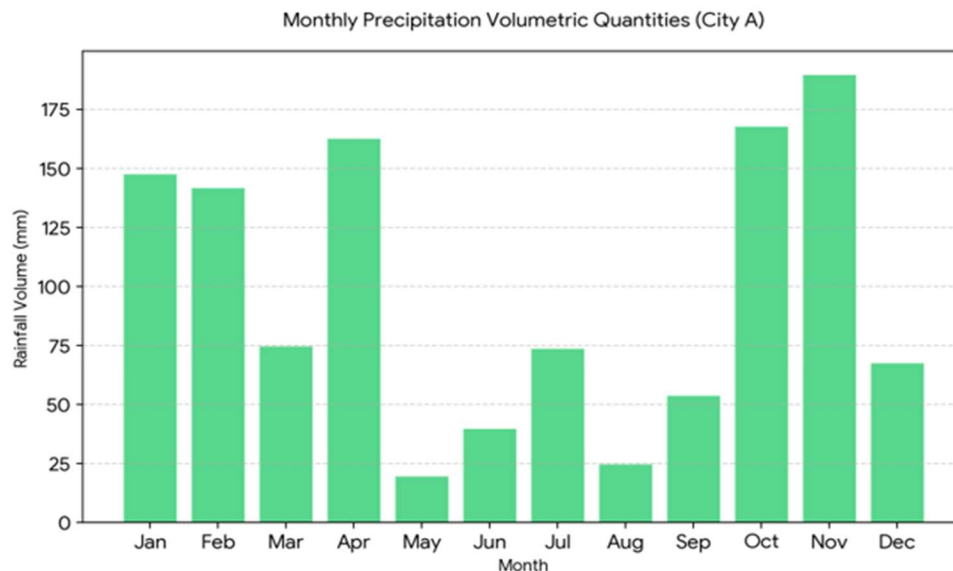
```
  geom_bar(stat = "identity", fill = "#2ecc71") +
```

```
  theme_minimal() +
```

```
  labs(title = "Monthly Precipitation Volumetric Quantities",
```

```
        x = "Month", y = "Rainfall Volume (mm)")
```

output:



Task 5: Bank Loan Risk Assessment

A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers. Recommend the visualization that should be included in a management dashboard. To safeguard institutional portfolios, a management dashboard must feature a Scatter Plot with custom color-mapped risk clusters or a clear Credit Score Histogram with superimposed risk bounds. Scatter plots help credit underwriters instantly evaluate multi-dimensional correlations (e.g., high loan amounts paired with dangerous low credit scores). This allows risk managers to pinpoint high-leverage outliers who deviate drastically from institutional risk policies, facilitating rapid visual audit trails.

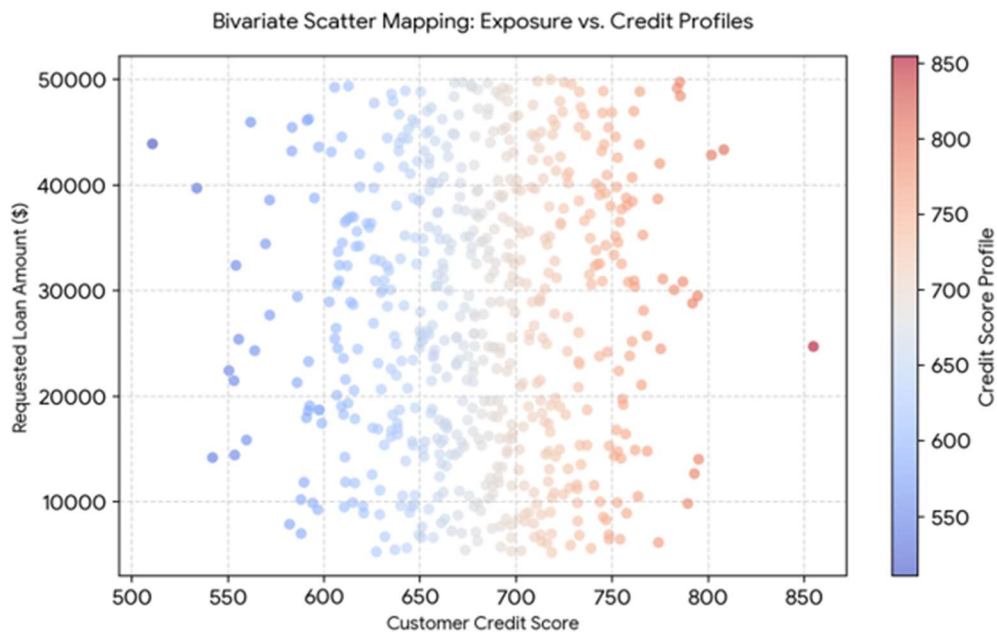
Python Implementation:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load credit risk assessment registry
loan_data = pd.read_csv("loan_risk.csv")
```

```
# Scatter plot representing risk patterns and bivariate distribution
plt.figure(figsize=(8, 5))
plt.scatter(loan_data['CreditScore'], loan_data['LoanAmount'],
            c=loan_data['CreditScore'], cmap='coolwarm', alpha=0.6)
plt.title('Bivariate Scatter Mapping: Exposure vs. Credit Profiles')
plt.xlabel('Customer Credit Score')
plt.ylabel('Requested Loan Amount ($)')
plt.tight_layout()
plt.show()
```

Output:



R Implementation:

```
library(ggplot2)

# Read loan portfolio evaluation data
risk_data <- read.csv("loan_risk.csv")
```

```
# Display population risk concentration via histogram
ggplot(risk_data, aes(x = CreditScore)) +
  geom_histogram(bins = 20, fill = "#9b59b6", color = "white") +
  theme_light() +
  labs(title = "Credit Score Distribution & Skewness Profiling",
       x = "Credit Score Metric", y = "Frequency Volume")
```

Output:

